

OMAR FARHAN

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SUMMARY

Data Science and Economics student at UC Davis (2027) working in AI engineering, data, and quantitative analysis. Now building Folio3's enterprise AI gateway on AWS and RoK, Apache ResilientDB's native token layer at ExpoLab. Founded a 6-person startup; shipped 100K+ record pipelines, 0.956 AUC models, and a 0.69 to 2.01 Sharpe trading engine.

EDUCATION

Double Major – BS, Data Science; BA, Economics; Technology Management Minor | University of California, Davis June 2027

EXPERIENCE

Folio3 Software

Remote (Belmont, CA)

AI Engineering Intern

June 2026 – Present

- Engineer the AWS build of Folio3's enterprise AI gateway (AI Guardian), extending a shipping Azure product to AWS in 5 weeks — routing Slack and Teams messages to guardrailed, retrieval-augmented Bedrock agents with input, output, and tool-loop controls, token quotas, and resumable serialized session state.
- Build SCIM synchronization and RBAC from Entra ID and Okta into a relational store driving per-department guardrail and quota policy, engineered to product targets of 45% lower AI spend and 60% faster retrieval; containerize Python services to Amazon ECS and Fargate via AWS CDK and CI/CD.

Apache ResilientDB (Incubating), ExpoLab — UC Davis

Davis, CA

Undergraduate Research Assistant

April 2026 – Present

- Build RoK, a native token layer for Apache ResilientDB (PBFT-based permissioned blockchain fabric) — balance management APIs and tooling in Python and C++ for account-based balances, cryptographic identity ownership, and transfers.
- Ship peer-reviewed contributions merged into the open-source Apache codebase, giving the fabric a standardized mechanism for transaction and execution costs and a base for gas accounting.

ZestOps

Lyon, France

Data Analyst Intern

June 2025 – September 2025

- Improved extraction accuracy of the multimodal vision model behind a receipt-capture product, where vendors photograph receipts for OCR parsing and cloud routing, cutting vendor correction time by 66%.
- Built Bronze-to-Gold data pipelines processing 100K+ records from APIs, web scraping, and internal sources using Python, SQL, AWS S3, and Snowflake, with JSON validation, record linkage, and cloud routing.

Shabah (Stealth FinTech Startup)

Riyadh, Saudi Arabia

Product Management Intern

August 2022 – April 2023

- Shaped the product roadmap for Saudi Arabia's first cryptocurrency payment platform alongside the founder, interviewing 30+ businesses to validate product-market fit and drive feature prioritization.
- Proposed and specified a Binance Wallet integration that expanded the platform beyond MetaMask into a core differentiator, and tested product workflows to refine the pre-launch user experience.
- Used Snapchat analytics to shape content strategy, driving the team's top post to 173K impressions.

TECHNICAL PROJECTS

Hala — Social Restaurant Discovery App

2025 – Present

- Founded and led engineering, product, and design teams building a React Native, TypeScript, and Supabase social media app that replaces inflated public reviews with your own circle, reaching 100+ beta testers across 5 Middle East cities.

ClockIt — Gesture-Controlled macOS Dictation App

2026

- Swift and SwiftPM macOS menu bar app that starts and stops on-device dictation from a hand gesture — AVFoundation capture feeding Vision hand-landmark tracking at 15 fps, driving a hold, release, and long-hold-to-latch state machine that held 92.66% gesture-detection accuracy across an 8-hour session; WhisperKit transcription and CGEvent Unicode insertion at the cursor, fully local with no network calls.

Statistical Arbitrage Engine

2026

- S&P 500 sector ETF pairs-trading engine (cointegration, Kalman-filtered spreads, HMM regime detection, 10-year walk-forward backtest) that lifted out-of-sample Sharpe from 0.69 to 2.01, win rate from 48% to 58%, and cut max drawdown 64%, shipped as a Streamlit dashboard.

NBA Injury Prediction — Machine Learning

2025

- Injury-risk classifiers trained on 11,000+ player-season records and 26 engineered workload and movement features (SMOTE, cross-validated Logistic Regression / Random Forest / XGBoost), reaching 0.956 AUC and 88% accuracy.

Steam Game Analytics Pipeline

2025

- Multi-stage ETL pipeline ingesting 1,000+ games from the Steam Web API, SteamSpy API, and web scraping into a normalized SQLite database, with Python/SQL analysis isolating the drivers of review performance.

LEADERSHIP

President, Muslim Tech Collaborative | **Vice President**, Data Science Club | **President**, Muslim Students' Association | **Head of Synergy**, Davis Economics Collective | **Executive Decision Maker**, ASUCD International Student Committee | **Vice President**, Amal House

SKILLS

Programming, ML & Data: Python, SQL, R, pandas, NumPy, scikit-learn, XGBoost, SMOTE, SHAP, GARCH, HMM, Kalman filter, feature engineering, model evaluation, ETL pipelines, record linkage, API integration, web scraping, Bronze/Silver/Gold architecture, AWS (S3, Bedrock, ECS/Fargate, CDK), Snowflake, Supabase, PostgreSQL, SQLite, Docker, CI/CD, RAG, LLM agents, statsmodels
Software, Web, Mobile & Viz: Matplotlib, Seaborn, Streamlit, Git, REST APIs, React Native, Flutter, TypeScript, Solidity, Web3.js, full-stack, backend, frontend, Expo, mobile development, data visualization, dashboards, API development, cloud deployment, Node.js, Kubernetes, Agile, Swift, macOS, computer vision, on-device ML

Languages: English (Native), Arabic (Native – Modern Standard & Gulf Dialects), Malayalam (Fluent), Urdu (Professional)